HIMANSHU SHARMA

Bhubaneswar, Odisha • himanshusharmazen@gmail.com • +91 9555839597 • GitHub • LinkedIn • Portfolio

EDUCATION

B.Tech in Computer Engineering — International Institute of Information Technology, Bhubaneswar 2023 – 2027
CGPA: 7.95 | Graduating 2027

Class 12 (ISC Board) — Dr. Virendra Swarup Memorial Public School, Kanpur
94%

TECHNICAL SKILLS

Languages: Python, SQL, C++, JavaScript (ES6+)

Data & SQL: SQL Query Design, SQL Validation & Optimization, SQLGlot/AST-Based Validation, Data Transformation, Structured Data Extraction, Data Pipeline Design

Databases: PostgreSQL, MySQL, SQLite, Supabase

LLM & AI Systems: LangGraph, LangChain, Groq API, Gemini API, Prompt Engineering, LLM-Based Evaluation

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Design, Operating Systems, DBMS, Computer Networks, Complexity Analysis

Web & Backend: React, Node.js, Express.js, REST APIs, Socket.io, WebRTC, HTML5, CSS3, TailwindCSS

Tools & Testing: Git, GitHub, Pytest

PROJECTS

Multi-Agent Text2SQL System — *Python, LangGraph, LangChain, Groq API, SQLGlot, Pytest*
GitHub

- Architected a stateful 7-node LangGraph pipeline (relevance → generation → validation → evaluation → feedback → execution → finalizer) that converts natural-language business questions into validated SQL queries through an iterative feedback loop
- Engineered a two-layer SQL validation system: deterministic AST-based validation (SQLGlot) enforcing syntax and schema correctness, paired with an LLM-based semantic evaluator to verify query intent against the business question
- Implemented tiered LLM compute via the Groq API, routing classification to an 8B model and complex SQL generation to a 70B model to balance cost and latency, with structured JSON outputs enforced through Pydantic
- Developed a comprehensive suite of 76 Pytest unit tests covering validators and routers; secured the execution layer with read-only SQLite constraints and resolved a nested subquery alias-resolution bug

CodeRace — *Real-Time Multiplayer Coding Race Platform — React, Node.js, Socket.io, Monaco Editor, OneCompiler API, Gemini API, Supabase*
GitHub | Live Demo

- Designed an event-driven, multi-tiered architecture with Socket.io to track and synchronize live match-state data (test cases passed, submission status, countdown) across many concurrent users in real time
- Built backend APIs integrating the OneCompiler API for multi-language code execution (Python, C++, Java) and Monaco Editor for a performant in-browser IDE
- Integrated the Gemini API into a prompt-engineered, penalty-based hint pipeline, balancing LLM-generated assistance against competitive fairness, with offline fallback logic

Stash-It — *Cross-Device File & Snippet Sharing Platform — React, Node.js, Express.js, Socket.io, WebRTC, Supabase Storage, TailwindCSS*
GitHub | Live Demo

- Built a full-stack file and snippet sharing platform combining Cloud Stash (Supabase Storage) with a peer-to-peer Direct Link mode via WebRTC
- Implemented Socket.io signaling with STUN/TURN servers for direct peer connections, enabling secure chunked transfer with real-time progress, speed, and ETA tracking
- Developed RESTful APIs for file management, unique share keys, expiration-based access control, and signed URLs

ACHIEVEMENTS & CERTIFICATIONS

- Solved 300+ Data Structures & Algorithms problems across LeetCode and GeeksforGeeks — arrays, graphs, dynamic programming, and trees
- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional — applied knowledge of LLMs, prompt engineering, and AI-powered application development
- Oracle Cloud Infrastructure 2025 Certified Foundations Associate — validated foundational knowledge of core OCI services, security & identity, billing, and governance
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate — foundational knowledge of AI/ML concepts, supervised & unsupervised learning, and generative AI on OCI
- HackerRank Software Engineer Role Certification — validated proficiency in problem-solving, data structures, and algorithms
- HackerRank SQL (Intermediate) Certification